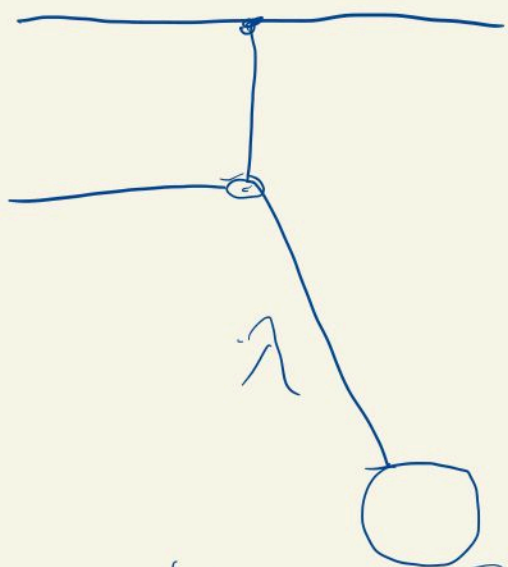




$$H(x, p, w(t)) = \frac{p^2}{2m} + \frac{1}{2} m w^2(t) x^2$$

$$\textcircled{1} \quad \frac{dH}{dt} = \frac{\partial H}{\partial x} \dot{x} + \frac{\partial H}{\partial p} \dot{p} + \frac{\partial H}{\partial t}$$

$$\textcircled{2} \quad \frac{\partial H}{\partial p} = \dot{x}, \quad \frac{\partial H}{\partial x} = -\dot{p}$$

$$\rightarrow \frac{dH}{dt} = \frac{\partial H}{\partial t}$$

(*) time scale for change >>

time scale for oscillation

$$(*) \gg \frac{2\pi}{\omega(t)} = T$$

$$\left| \left(\frac{2\pi}{\omega} \right) \cdot \frac{d\omega}{dt} \right| \ll \omega$$

$$\Leftrightarrow \left| \frac{\dot{\omega}}{\omega^2} \right| \ll 1$$

$$I = \frac{H}{\omega} \rightarrow \frac{\partial I}{\partial t}$$

$$= \frac{\dot{\omega} H - \omega \dot{H}}{\omega^2} = \left(\frac{\dot{\omega}}{\omega^2} \right) (V(t) - K(t))$$

$$\overline{V(t)} = \overline{K(t)}$$

$$I(T+t) - I(t) = \int_0^T \frac{\partial I}{\partial t} dt$$

$$\approx \frac{\hbar}{\omega^2} (\overline{V} - \overline{K})$$

$\rightarrow$ small.

QM: Quantum number of system does not change

Circuit (based) Computation:

$$U = U_L \dots U_1 \quad U_i = \text{unitary}$$

$$|\psi_{\text{fin}}\rangle = U |\psi_{\text{ini}}\rangle$$

$$H(t)$$

$$U(t, t_{ini})$$

$$i \frac{\partial U}{\partial t}(t) = H(t) U(t, t_{ini})$$

$$|| \Psi_{fin} \rangle = U(t_{fin}, t_{ini}) | \Psi_{ini} \rangle$$

$$T = t_{fin} - t_{ini}$$

$|\Psi_{fin}\rangle$: Eigenstate of

$$H_{fin}$$

$$H: [t_{ini}, t_{fin}] \rightarrow B_{sa}(H)$$

$$t \mapsto H(t)$$

$$H(t_{ini}) = H_{ini}$$

$$H(t_{fin}) = H_{fin}$$

$$; \frac{d}{dt} U(t, t_{ini}) = H(t) U(t, t_{ini})$$

$$U(t_{ini}, t_{ini}) = \underline{1},$$

$$H_T: [0, 1] \rightarrow B_{sa}(H)$$

$$H_T(S) = H(t_{ini} + ST)$$

$$t = t_{ini} + ST, \quad S = \frac{t - t_{ini}}{T}$$

$$H_T(0) = H(t_{ini}) = H_{ini}$$

$$H_T(1) = H(t_{fin}) = H_{fin}$$

$$i \frac{d}{ds} U_T(s) = T H_T(s) U_T(s)$$

$$U_T(0) = 1$$

$$U_T(1) = U(t_{fin}, t_{ini})$$

$E_j(s)$: j th eigenvalue of $T H_T(s)$

$$j \in I \quad |I| = \infty$$

$$I \subset \mathbb{N}_0$$

$$|\Phi_{j,\alpha}(s)\rangle \in \mathcal{H}$$

$$\alpha \in \{1, \dots, d_j\}, d_j < \infty$$

$$\langle \Phi_{j,\alpha} | \Phi_{k,\beta} \rangle = \delta_{j,k} \delta_{\alpha,\beta}$$

$j \neq k$

$$H_T(s) |\Phi_{j,\alpha}(s)\rangle = E_j(s) |\Phi_{j,\alpha}(s)\rangle$$

$$E_j(s) : [0, \infty) \rightarrow \mathbb{R}$$

instantaneous eigenvalues

$$|\Phi_{j,\alpha}(s)\rangle \neq U_T(s) |\Phi_{j,\alpha}(0)\rangle$$

Adiabatic Assumption (AA)

$$H_T : [0, 1] \rightarrow B_{sa}(H)$$

(i) $\ddot{H}_T$ exist for every $s \in [0, 1]$

(ii) H_T has purely discrete spectrum

(iii) $E_i(s) \in H_T(s)$

$$\underline{E_0}(s) < E_1(s) < \dots < \underline{E_i}(s)$$

$$|\Phi\rangle, |\Psi\rangle \in \mathcal{H}, \quad \varepsilon \in \mathbb{R}$$

$$\| |\Phi\rangle - |\Psi\rangle \| \leq \varepsilon$$

$$\rightarrow \left| \langle \Phi | \Psi \rangle \right|^2 \geq 1 - \varepsilon^2$$

pf):

$$2 - 2\operatorname{Re} \langle \Phi | \Psi \rangle \leq \varepsilon^2$$

$$\rightarrow \operatorname{Re} \langle \Phi | \Psi \rangle \geq 1 - \frac{\varepsilon^2}{2}$$

$$\rightarrow |\langle \Phi | \Psi \rangle|^2 \geq \operatorname{Re} \langle \Phi | \Psi \rangle^2$$

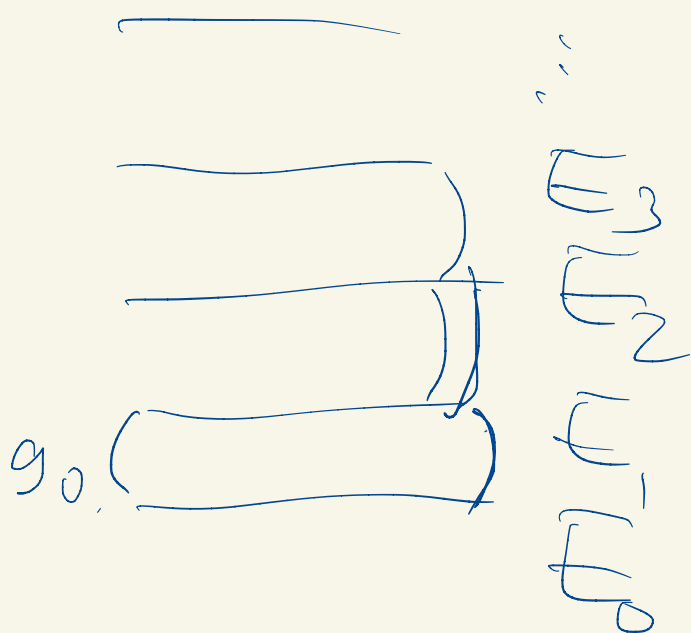
$$\geq \left(1 - \frac{\varepsilon^2}{2}\right)^2 \geq 1 - \varepsilon^2$$

Let

$$H_T: [0,1] \rightarrow B \text{ sa } (H)$$

$$g_j: [0,1] \rightarrow \mathbb{R}$$

$$g_j(s) = \min_{K \in \mathcal{K}} \{ |E_K(s) - G(s)| \}$$



(Def 6.7)

$$\underline{H_{A,j}} : [0,1] \rightarrow B_{sa}(H)$$

$$\underline{H_{A,j}(s) = T_{H_T}(s) + i [P_j(s), E_j(s)]}$$

where

$P_j(s)$: projection onto the
eigenspace of $E_j(s)$

$$i \dot{U}_{A,j} = H_{A,j} U_{A,j}$$

$$U_{A,j}(0) = 1$$

$$\Rightarrow U_{A,j}(s)^* P_j(s) U_{A,j}(s) = P_j(s)$$

① $s \approx 0$: easy

② $\frac{d}{ds} (U_{A,j}(s)^* P_j(s) U_{A,j}(s))$

Quantum Adiabatic Thm

$\rightarrow H_\tau = (AA)$

$U_{A,j}, H_{A,j}, P_j, Q_j$

$\rightarrow \left\| (U_{A,j}(s) - U_\tau(s)) P_j(0) \right\| \leq \frac{C_j(s)}{T}$

$$C_j(s) = \frac{\|\dot{H}_T(s)\|}{g_j(s)^2} + \frac{\|\dot{H}_T(0)\|}{g_j(0)^2} + \int_0^s \left(\frac{\|\ddot{H}_T(u)\|}{g_j(u)^2} + \frac{\|\dot{H}_T(u)\|^2}{g_j(u)^3} \right) du$$

Cor 8.3

If H_T satisfies (AA)

$$E_j = \frac{C_j(1)}{T}$$

$|E_j\rangle$: initial eigenstate
of $E_j(0)$ of H_{in_i}

↓ after time evolution

$$\| \underbrace{P_j(t) U_T(t)}_{\downarrow} |E_j\rangle \|$$

projection to the eigenstate of $E_j(t)$

$$\| P_j(t) U_T(t) |E_j\rangle \| \geq 1 - \epsilon_j^2$$

$$|\Phi(s)\rangle = U_{A_j}(s) P_j(0) |E_j\rangle$$

$$|\Phi(s)\rangle = U_T(s) P_j(0) |E_j\rangle$$

$$\| |\Phi(s)\rangle - |\bar{\Phi}(s)\rangle \|$$

$$= \| (U_{A_j}(s) - U_T(s)) P_j(0) |\Xi_j\rangle \|$$

$$\leq \| (U_{A_j}(s) - U_T(s)) P_j(0) \|$$

$$\leq \frac{C_j(s)}{T} = \epsilon_j(s)$$

$$|\langle \Phi(s) | \bar{\Phi}(s) \rangle|^2 \geq (1 - \epsilon_j(s))^2$$

What we want: $\| P_j(0) U_T(1) \Xi_j \|$

→ So we have to use the fact

$$U_{A_j}^*(s) P_j(s) U_{A_j}(s) = P_j(0)$$

$$\| P_j(s) U_T(s) \Xi_j \| ^2$$

$$= \| U_{A,j}(s)^* P_j(s) U_T(s) \Xi_j \| ^2$$

$$= \| P_j(0) U_{A,j}^* U_T(s) \Xi_j \| ^2$$

Ξ_j : eigenstate of $\Xi(0)$

P_j : projection to eigenspace of Ξ_j

$$\rightarrow \| P_j(0) \Omega \| ^2 \geq | \langle \Xi_j | \Omega \rangle |^2$$

projection onto

subspace $\{ \pm \Xi_j \}$

$$\rightarrow \|P_j(S)U_T(S)\Xi_j\|^2$$

$$\geq |\langle \Xi_j | U_{A_j}(S)^* U_T(S) \Xi_j \rangle|^2$$

$$= |\langle U_{A_j}(S) \Xi_j | U_T(S) \Xi_j \rangle|^2$$

$$\leq |\langle \mathbb{I}(S) | \mathbb{I}(S) \rangle|^2$$

$$\geq 1 - \epsilon_j(S)^2$$

$$S=1$$

Cor 8.4

$P_{\min} > 0$.

// $P_{\min}$

$$\text{prob} \geq 1 - \left(\frac{\epsilon(1)}{T} \right)^2$$

$$\Leftrightarrow T \geq \frac{C_i(1)}{\sqrt{1 - P_{\min}}}$$

We cannot make $T \rightarrow \infty$

because of decoherence effect.

Generic Adiabatic Algorithm

Input: A problem s.t. the solution can be obtained through knowledge of

$$|s\rangle \in H$$

- Find H_{fin} s.t. $|s\rangle$ is an eigenstate for some $E_{\text{fin},j}$

Typically j is chosen as $j=0$.

$$S_j(1) = S_j(1; \underline{g_j}, \dot{H}_T, \ddot{H}_T)$$

we have to make g_j large.

Typically $\sum_{j=0}^N$

② Find H_{in_j} , eigenstate $|\Phi_{in_j}\rangle$

can easily be prepared

e.g. $|x\rangle = \underbrace{|1 \dots 1\rangle}_N$

③ ^{find} $H(t)$ s.t.

$$H(t_{in}) = H_{in_j}$$

$$H(t_{fin}) = H_{fin}$$

$$H_\tau(S) = H(t_{ini} + ST). \quad T = T_{function}$$

satisfy (AA)

④ $|\Phi_{ini,i}\rangle$ (prepare)

⑤ Evolve the system
by $U(\tau, t_{ini})$
until $t = t_{fin}$

⑥ $U(t_{fin}, t_{ini}) |\Phi_{ini,i}\rangle$

observation
→

$|F_S\rangle$

with

probability $\geq 1 - \left(\frac{\epsilon_{SU}}{T}\right)^2$

but... we have to find

$$C_j(1) = g_j, \quad \dot{H}_T, \ddot{H}_T$$

$$H_T(s) = (1 - f(s)) H_{ini} + (f(s)) H_{fin}$$

$$\begin{array}{ccc} & f(s) & \\ & \bullet & \\ \text{H}_{ini} & \text{---} & \text{H}_{fin} \end{array}$$

$$f(0) = 0, \quad f(1) = 1$$

$$f' > 0, \quad f'' \text{ exists}$$

$$C_j(s) = \| H_{fin} - H_{ini} \|$$

$\times$

$$\left[\frac{\dot{f}(s)}{g_i(s)^2} + \frac{\dot{f}(0)}{g_i(0)^2} \right.$$

$$\left. + \int_0^s \frac{\|\ddot{f}(u)\|}{g_i(u)^2} + (0) \|H_{fin} - H_{ini}\| \frac{\dot{f}^2}{g_i^3} du \right]$$

QUBO: Quadratic Unconstrained
Binary Optimization

$$Q \in M^{n \times n}(\mathbb{R})$$

$$B: \{0, 1\}^n \rightarrow \mathbb{R}$$

$$(x_0, \dots, x_{n-1}) \mapsto \sum_{i,j} x_i Q_{ij} x_j$$

we choose

$$H_{ini} = \sum_{j=0}^{n-1} \Sigma_z^j$$

$$H_{fin} = \sum_{j=0}^{n-1} K_j \Sigma_z^j +$$

$$\sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} J_{ij} \Sigma_z^i \Sigma_z^j + C \mathbb{I}^{\otimes n}$$

where

$$\Sigma_z^j = \mathbb{1}^{\otimes n-1-j} \otimes \sigma_z \otimes \mathbb{1}^{\otimes j}$$

$$J_{ij} = \frac{1}{4} Q_{ij} \quad \text{for } i \neq j$$

$$K_j = -\frac{1}{4} \sum_{\substack{i=0 \\ i \neq j}}^{n-1} (Q_{ij} + Q_{ji}) - \frac{1}{2} Q_{jj}$$

$$C = \frac{1}{4} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} Q_{ij} + \frac{1}{2} \sum_{i=0}^{n-1} Q_{ii}$$

$$\sigma_z |0\rangle = |0\rangle$$

$$\sigma_z |1\rangle = -|1\rangle$$

$$E_{in_i, j} = -n + 2j \quad j=0, \dots, n$$

with degeneracy $\binom{n}{j}$

$$\rightarrow E_{in_i, 0}, E_{in_i, n}$$



$$| \uparrow_0 \rangle = | 1 \dots 1 \rangle = | 2^n - 1 \rangle$$

$$H_{fin} |x\rangle = \beta(x) |x\rangle$$

$$|x\rangle = |x_n \dots x_0\rangle \in H^{\otimes n}$$

→ Finding a minimum point x is just finding a minimum eigenvalue of H_{fin}

$$= E_{fin,0}$$

→ We observe $E_{fin,0}$ with prob. $\geq 1 - \epsilon_i^2$

8.4. Adiabatic Quantum search

$$x \in S \subset \{0, \dots, N-1\}$$

$$N = 2^n$$

$$|\Phi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

: eigenstate for

$$H_{\text{ini}} = \mathbb{I} - |\Phi_0\rangle\langle\Phi_0|$$

$$E_{\text{ini}} = 0$$

$$|\Phi_1\rangle \perp |\Phi_0\rangle \rightarrow$$

$$H_{\text{ini}} |\Phi_1\rangle = |\Phi_1\rangle$$

eigenstates for E_{ini}

$$H_{fin} = 1 - P_S$$

where

$$P_S = \sum_{x \in S} |x\rangle \langle x|$$

$$H(t) = H_T \left(\frac{t - t_{ini}}{\tau} \right)$$

$$H_T(s) = (1 - f(s)) H_{ini} + f(s) H_{fin}$$

we want: final states

$$|\overline{\Psi}_{fin}\rangle \text{ s.t.}$$

$$H_{fin} |\Phi_{fin}\rangle = 0$$

↓

$$1 - P_S$$

$\Rightarrow |\Phi_{fin}\rangle$ is contained in
 $\text{span} \{x: x \in S\}$

Thm 8.11

① $H_T(0)$ has eigenvalue

$$\begin{cases} E_-(0) = 0 \\ E_+(0) = E_1(0) = E_2(0) = \end{cases}$$

non-degen.
 $\rightarrow$

↓
 degenerate with order
 1-1

② $H_T(s), \quad s \in (0, 1),$

has eigenvalues

$$\begin{aligned}
 & \left\{ \begin{aligned} E_-(s) &= \frac{1}{2} - \frac{1}{2} \sqrt{\tilde{m} + 4(1-\tilde{m})(f(s) - \frac{1}{2})^2} \\ E_+(s) &= 1 - f(s) \\ E_2(s) &= 1 \end{aligned} \right. \\
 & m \leftarrow E_-(s) \\
 & N-m-1 \leftarrow E_2(s) = 1 \\
 & m = |S| \qquad \tilde{m} = \frac{m}{N}
 \end{aligned}$$

③ $H_\tau(1)$ has eigenvalues

$$E_-(1) = E_1(1) = 0, \quad : m\text{-fold}$$

$$E_+(1) = E_2(1) = 1 \quad : N-m \text{ fold.}$$

$$\begin{aligned}
 H_\tau(s) &= (1-f(s)) \underbrace{H_{in}}_{|-|\Phi_0\rangle\langle\bar{\Phi}_0|} + f(s) H_{fin} \\
 &\qquad\qquad\qquad \downarrow \\
 &\qquad\qquad\qquad 1 - P_S
 \end{aligned}$$

6.9
Grover's search algorithm ?

efficiency of QAC, Grover's
are the same.

$$f(S) = S \rightarrow \text{QAC gives}$$
$$T \in O\left(\frac{N}{m}\right)$$

minimum time to wait
to achieve $P_{\min}$ to solve
 $x \in S$

Thm 8.16

If we choose f as a solution of

$$\begin{cases} \dot{f}(s) = K_b g(f(s))^b \\ f(0) = 0 \end{cases}$$

where $K_b = \int_0^1 g(u)^{-b} du$

$g(u)$: gap ftn of E_-

$$\rightarrow g(u) = \sqrt{\hat{m} + 4(1-\hat{m})\left(u - \frac{1}{2}\right)^2}$$

$$1 < b < 2$$

$$\rightarrow T \in O\left(\sqrt{\frac{N}{m}}\right)$$

$\approx$ Grover's algorithm

(circuit-based)

Q: Circuit based & QAC
Efficiency ?

8-5, 8-6 $\rightarrow$ Same efficiency
